Hasan Mohd Hussain

Software Engineer | *Python Backend* | *Full-Stack Development*
Mumbai | 7400341331 | hasan98209@gmail.com | Portfolio | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Software Engineer with an **MS in Computer Engineering**, building **Python backend services**, **RESTful APIs** and **full-stack applications** on **AWS**. Delivered **95% faster QA** and **20% faster API responses** through **test automation** and backend optimization. Skilled in **FastAPI, Flask, SQL/NoSQL, Docker and CI/CD**.

EDUCATION

University of Texas at Dallas — **Richardson, Texas, USA**
Master of Science in Computer Engineering — Aug 2024 - Aug 2026
GPA: 3.74/4.0 | *Relevant Coursework: NLP, Machine Learning, Computer Architecture, Embedded Systems*

University of Mumbai — **Mumbai, Maharashtra, India**
Bachelor of Engineering in Computer Engineering — Aug 2017 - Jul 2021
GPA: 7.88/10.0 | *Relevant Coursework: Data Structures, Database Management Systems*

SKILLS

Languages: Python, Java, C++, C#, JavaScript, TypeScript, SQL
Backend & APIs: FastAPI, Flask, Django, Node.js, Express, RESTful API design, WebSockets, authentication
Frontend & Databases: React, HTML5, CSS3, MongoDB, PostgreSQL, MySQL, SQLite, schema design, indexing
Cloud & DevOps: AWS (ECS Fargate, ECR, EC2, S3), Docker, Kubernetes, CI/CD, Git, Linux
Testing & Engineering: PyTest, unit testing, Playwright, Postman, OOP, DSA, SDLC, Agile/Scrum
AI Integration: LLM APIs, RAG, LangChain, LangGraph, OpenAI API, Azure OpenAI, Gemini API, Ollama, Claude Code, Claude Cowork

WORK EXPERIENCE

Mphasis — **New York, United States**
Machine Learning Engineer — May 2025 - Aug 2025, Jun 2026 - Aug 2026
- Engineered an **end-to-end QA automation agent** with **Python, Flask and Playwright** for a production mortgage dashboard.
- Integrated **multi-provider LLM grading**, cutting per-case QA runtime by **95%** with reproducible **PDF/Excel reports**.
- Built a **synthetic test-data generator** for **5 financial document types**, scaling coverage to **200 test cases per run**.
- Delivered a **PDF RAG service** using **ChromaDB, FastAPI and Azure OpenAI**, improving retrieval by **30%**.
- Optimized **Python backend logic** for **FastAPI, Docker and AWS** applications, improving **API response time by 20%**.

Dot9 Games — **Mumbai, India**
Game Developer Intern — Apr 2024 - Aug 2024
- Built gameplay systems in **Unity** and **C#** from design documents, shipping features that raised **user engagement by 25%**.
- Ran **POC testing** and **QA** on new game designs and optimized performance to a stable **60 FPS**, managing code with **Git**.

Shree Infotech — **Mumbai, India**
Software Engineer — Apr 2023 - Mar 2024
- Developed **Python/Flask backend services** and **REST APIs** on **Docker**, automating workflows and saving **10+ hours/week**.
- Designed **MongoDB** schemas and indexes for **fast query retrieval**, and wrote **unit tests** (PyTest) to keep core services reliable.

Nilee Games — **Mumbai, India**
Game Developer Intern — Nov 2022 - Mar 2023
- Built a **2D physics-based driving game** in **Unity/C#** within **two weeks**, contributing to **500K downloads** in its first month.
- Raised **user engagement by 30%** through gameplay tuning and **QA testing** on Android devices, using **Git** for source control.

PROJECTS

JobQuest.AI | *FastAPI, SQLite, JavaScript, Multi-LLM*
- Built a **full-stack job-search application** with **FastAPI**, **SQLite** and a **multi-provider LLM integration layer**.
- Delivered **application tracking**, **Chart.js analytics** and **Gmail IMAP/SMTP sync**; packaged with **PyInstaller** and **53 tests**.

Recipe Manager | *Flask, MongoDB, Docker, AWS ECS Fargate*
- Built **authenticated CRUD workflows** with **Flask/MongoDB** and optimized indexes for **40% faster queries**.
- Deployed **Docker containers** through **Amazon ECR** to **AWS ECS Fargate**, using **environment-based configuration**.

Keep Notes | *React, Express, Node.js, MongoDB Atlas*
- Built a **MERN full-stack application** with **REST APIs**, **React** components and **MongoDB Atlas** persistence.
- Implemented **live search**, **pinned notes**, **responsive layouts** and **Framer Motion** animations.

CERTIFICATIONS

Anthropic: Claude Code in Action, Claude Code 101, Claude 101 | **Udemy:** SQL Bootcamp | **DeepLearning.AI:** AI For Everyone
IBM: Machine Learning with Python, Building AI Agents and Agentic Workflows | **Meta:** Back-End Developer Professional Certificate